

Data Science

Andrea Savastano

19 febbraio 2026

Indice

1	PCA: Principal Component Analysis
---	-----------------------------------

4

prova

1 PCA: Principal Component Analysis

$$VAR[X_k] = \frac{1}{N-1} \sum_{i=1}^N (x_{ik})^2$$

$$VAR[T_k] = \frac{1}{N-1} \sum_{i=1}^N (t_{ik})^2 = \frac{1}{N-1} \sum_{i=1}^N \left(\sum_{j=1}^p x_{ij} w_{jk} \right)^2 = \frac{1}{N-1} \|X w^{(k)}\|^2$$

$$PVE = \frac{\sum_{k=1}^m VAR[T_k]}{\sum_{k=1}^p VAR[X_k]} = \frac{\sum_{k=1}^m \left[\frac{1}{N-1} \sum_{i=1}^N \left(\sum_{j=1}^p x_{ij} w_{jk} \right)^2 \right]}{\sum_{k=1}^p \frac{1}{N-1} \sum_{i=1}^N x_{ik}^2}$$

Elenco delle figure